State Methodology (10/22)

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22 October 2026

This appendix documents the state-specific changes to the Budget Lab's ECEC model methodology. I use its definitions, notation, and model structure throughout.

I process and calibrate the model nationally for 2019, then adjust supply and re-anchor demand for each geography. State simulations cover children ages 0–4 in 50 states, DC, and New York City (NYC). I exclude NYC's separate run from national totals to avoid double counting with New York State.

Inputs and fitted adjustments

I retain the national model's joint employment and care choices, utility function, and equilibrium conditions. The state extension changes inputs and baseline adjustments:

- **Shared parameters.** I estimate the care, employment, and price-wedge forests nationally. All geographies share β, ρ, production technology, and care-worker labor-supply elasticities.
- **Local inputs.** I supply annual child-hours in six care types, wages and wage growth, staffing requirements, non-labor costs, and caregiver employment paths.
- **Fitted adjustments.** p_0 denotes predicted national baseline choice probabilities. Alpha contains the choice-specific preference terms. I first adjust alpha to preserve p_0 at local prices, then fit care-type shifts δ and caregiver employment shifts to local baseline targets. These shifts enter utility; they are not prices or elasticities.
- **Outcomes.** The model determines equilibrium prices and wages, the split between low- and high-price centers, and policy responses. I do not impose local targets on policy outcomes.

1 Income ranks and calibration

I retain the national NSECE-to-ACS estimation framework and add state and county identifiers to ACS households.

The national model ranks families on nominal income. For state runs, I divide 2019 income by the 2019 BEA all-items regional price parity divided by 100. I then apply national percentile cutpoints built separately in NSECE and ACS.

I keep the 2019 training ranks fixed. All-items parity adjusts income for general purchasing power; childcare affordability enters through care prices and household payments.

ACS households use their state's parity. NSECE identifies only the Census region, so NSECE households use the child-weighted mean parity of the states in their region. NYC uses the New York metro parity, 114.9, and gets its own ranks and alpha.

I retain the national forests, utility specification, and elasticity targets. I recalibrate β and ρ nationally after the income adjustment; states share these parameters.

1.1 Calibration results

The price-adjusted calibration completed on 19 October 2026 gives utility scale $\beta = 0.0003405$ and curvature $\rho = 0.0262383$, shared by one-child and two-or-more-child parent units.

Aggregate elasticity	Target	Achieved
Employment with respect to paid care cost	−0.15	−0.1500
Work hours with respect to after-tax income	−0.05	−0.0500
Paid care hours with respect to paid care cost	—	−1.5137
Employment with respect to caregiver wages	—	1.5636

The first two rows are calibration targets; the remaining rows are model-implied responses, not additional targets. All four elasticities aggregate with parent-unit weights. The care-cost experiment raises paid care costs by 1%; the income experiment adds 1% of each family's expected after-tax income as a lump-sum transfer.

1.2 Center price tiers

The national model separates low- and high-price household center care at $4.75 an hour. For state runs, I classify posted prices and household payments with a threshold, which changes by region, urban density, and age band. A price at or above the threshold is high-price. The tiers use price as a proxy for quality.

The regional samples for rural and moderate density are small. I start with each region's high-density median. I scale it by the national price ratio between density groups:

$$\underset{\text{(region, urban density, age band)}}{\text{threshold}} = \underset{\text{high urban density, age band}}{\text{regional median price,}} \times \frac{\underset{\text{this urban density, age band}}{\text{national median price,}}}{\underset{\text{high urban density, age band}}{\text{national median price,}}} \tag{1}$$

I calculate medians over positive prices. The weight for each center is its survey weight times its full-time-equivalent enrollment. The posted price is the weekly rate divided by 40 hours. A household's price is its weekly payment divided by its weekly hours. The age bands are 0–1, 2–3, and 4–5.

Urban density is the NSECE community urban density. It has three levels:

- High: 85% or more of the nearby population is urban.
- Moderate: 30% to under 85% is urban.
- Rural: under 30% is urban.

A center uses its own level, while a family uses the level of its household. Each regional cell has 5–35 surveyed rural centers and 28–134 moderate-density centers per age band (unweighted counts of centers). I assume that the national ratio holds in every region. The care forest has no urban density feature.

2 Demand targets for each geography

A state's households have national preferences. I re-anchor each geography to its own level of care. I target annual hours by care type, an additional step in the state extension.

2.1 Children in regular care

$$\begin{matrix}\text{children in}\\ \text{regular care}\end{matrix} = \begin{matrix}\text{population}\\ \text{ages 0–4}\end{matrix} \times \text{NSCH rate} \times \begin{matrix}\text{five-hour}\\ \text{factor}\end{matrix} \tag{2}$$

The population is the Census 2019 county estimate for ages 0–4.

The National Survey of Children's Health (NSCH) rate is the share of children with at least 10 weekly hours of non-parental care. I pool the rate over 2016–2019 in each state. NSCH does not identify NYC. NYC uses New York's rate.

Regular care in NSECE starts at 5 hours a week, not 10. I scale the rate by one national five-hour factor. The factor is NSECE children in regular care divided by NSECE children in care 10 or more hours a week. Both counts use NSECE child weights. Both counts cover children ages 0–4 and exclude school programs. The factor is 1.098.

For the 50 states and DC, the result is 11.6 million children in regular care (weighted). This is 59.2% of the estimated population of 19.6 million children ages 0–4.

2.2 Unpaid center children

Unpaid Center-Based care takes each geography's own program counts:

1. Pre-K is the larger of public-school pre-K from the Common Core of Data and an all-settings count. The all-settings count is Table 2 of the NIEER State of Preschool yearbook for a state, and the city's own count for NYC.
2. Funded Head Start and Early Head Start slots convert to enrolled children with the Office of Head Start enrollment ratios from the 2018–19 Program Information Report.
3. I remove home-based and family child care options, children age 5 and older, and Head Start children that pre-K already counts. I use one pooled Head Start overlap share: the slot-weighted mean across the 18 geographies with NIEER estimates.

I count adjusted pre-K and Head Start enrollment as unpaid-center children. This count can be too high, for two reasons. Children who also use paid centers get the paid-center label. Some public programs charge families. The overlap adjustment removes children that both pre-K and Head Start count. It does not remove overlap with other care.

2.3 Other care types

I subtract unpaid-center children from total children in regular care. I divide the remainder across the other five types with child-weighted p_0 shares in each geography. The same probabilities give the part-time share of each type. These are predicted shares, weighted by children; they are not direct state survey estimates. The mix shows local household characteristics under the national model.

2.4 Hours and re-anchoring

$$\text{annual hours of type} = \text{children in type} \times \Big(\text{part-time share} \times 500 + \text{full-time share} \times 2{,}000\Big) \quad (3)$$

I target paid center, unpaid center, paid home, unpaid home, other paid, and other unpaid care. The paid-center target combines low- and high-price centers. Part-time and full-time care represent 10 and 40 hours a week for 50 weeks.

At simulation start, I adjust alpha in two steps:

1. I express the baseline preferences at local 2019 prices. A lower care price raises the financial benefit of paid care, so I offset that initial gain in alpha. For example, a family with a 60% predicted probability of paid care keeps that probability when I first substitute local prices. I then fit the local targets in step 2. This defines the starting point; after these adjustments, policy-induced price changes affect choices with alpha held fixed.

2. I solve one constant δ for each care type. Every household in the geography shares the same δ. The δ values make base-year hours match the local targets. Parent Only has $\delta = 0$. A household's alpha for a choice moves by the sum of δ over the care types of its children. The two center tiers share one δ. The model sets the split between them.

   In the same solve, I add one shift on working choices for each caregiver group. The groups are sex by four age bands. The shift makes each group's 2019 employment rate equal its employment-forest rate. The run stops if the targets lack support or the solve does not converge.

The joint solve matches care hours and employment rates; child counts and part-time shares remain model outcomes. Yearly employment targets grow from these matched base-year rates.

Summing the 50 states and DC, the weighted annual child-hour targets exceed the price-adjusted national calibration by 9.0–9.5% in five care types and 22.7% in unpaid centers. NYC is excluded from both sums. The reference is the national calibration before local re-anchoring. The local targets combine survey and program counts, so they need not reproduce the national model's care totals. I retain the local targets and report the gaps.

3 Supply and employment for each geography

I anchor each group's base-year labor supply to the labor required for locally re-anchored care hours. I adjust wages, staffing requirements, and residual revenue by geography.

3.1 Wages

I scale each group's national base wage by its state-to-national wage index ratio. Workers without and with a bachelor's degree have national base wages of $12.06 and $19.70 an hour. The wage index mixes May 2019 hourly medians from the BLS Occupational Employment Statistics for childcare workers, preschool teachers, and preschool and childcare administrators. The mix uses the national ACS share of each group's workers in each occupation. NYC uses the New York metro area.

For later years, I multiply the 2019 wage by national hourly-wage growth and the change in the geography's local-to-national wage ratio. The annual ratios use 2019–2025 BLS data and fixed ACS occupation weights. Within each geography and education group, I use the same occupation basket in every year, dropping suppressed occupations from both local and national wages. I hold the geographic ratio fixed after 2025; national wage growth continues. The 2021 BLS methodology change limits comparability across years.

3.2 Staffing

The staffing index is the state's required staff per child divided by the national average. The national average weights states by children ages 0–4. I start with the 2017 Child Care Licensing Study (ICPSR 37700) and replace records with reviewed rules applicable on 1 October 2019.

For centers, required staff per child is the average of 1 over the child-to-staff ratio at five ages: 11 months, 18 months, 35 months, 3 years, and 4 years. NSECE 2019 center enrollment at those ages gives the weights. For homes, I use the inverse of the selected family-home capacity proxy. I distinguish total license capacity from children per adult and record age-composition limits and assistant requirements. Some rules permit several configurations, so a single home value requires an explicit assumption. Where family homes are not licensed, I use a reviewed state certification rule when it provides a comparable one-adult preschool capacity. I label that proxy separately from a licensing requirement. A state with neither input uses an index of 1. NYC centers use the June 2019 city Health Code. Idaho's center ratios come from its statutory points rule.

I multiply national labor requirements per child-hour by the center index for all three center sectors. I multiply them by the home index for paid homes. Each index applies equally to both worker groups. Legal limits do not measure actual staffing. I use them because they provide a consistent basis for geographic comparisons. Certification proxies and unresolved inputs remain identified in the audit.

3.3 Non-labor costs

The national residual-revenue amounts are $0.51 for low-price centers, $4.61 for high-price centers, and $2.83 for paid homes.

I scale each sector's national non-labor residual by the geography's 2019 BEA housing-services price parity divided by 100. NYC uses the metro value. Unpaid centers keep a zero residual. The factor goes from 0.55 to 1.70 across the 52 geographies. Housing-services prices proxy geographic differences in non-labor costs and provider margins. I keep these geographic factors fixed and grow each residual with the CPI, as in the national model.

I compare model prices with the National Database of Childcare Prices (NDCP) in 46 states. I compare two versions of the model price: one with the residual scaled by the housing parity, and one with the national residual, as in the national model. The housing scaling raises the unweighted Pearson correlation from 0.83 to 0.89. It lowers the weekly root mean squared error from $41.6 to $34.5. The ratio of the child-weighted mean model price to the child-weighted mean NDCP price rises from 1.09 to 1.12.

Restricting the comparison to counties with unimputed prices in every age group leaves only four states. In that small sensitivity sample, housing scaling raises weekly error from $48.8 to $51.6. The stronger fit above therefore applies to the broader benchmark, which includes imputed prices.

The comparison predates the latest staffing additions and will be refreshed after their rebuild. Applying the annual wage paths and CPI growth gives correlations of 0.90, 0.89, and 0.90 in 2020–2022; weekly errors rise to $39.0, $41.3, and $47.1. These later comparisons cover 46, 41, and 45 states, respectively, with 2019 care-mix weights held fixed. Better geographic fit does not establish accurate price growth or levels.

The comparison uses 2019 base supply prices converted to 40-hour weeks, with fixed model enrollment weights across price tiers and age groups. NDCP prices are means of county medians, weighted by children ages 0–4 and including imputed prices. The sources' age bands and full-time definitions do not exactly align:

- Age bands. NDCP infant, toddler, and preschool prices cover 0–23, 24–35, and 36–54 months. I match them to model ages 0–1, 2, and 3–4. Model age 4 includes children aged 55–59 months, which NDCP does not price.
- Full time. The model week is 40 hours. NDCP has no single full-time definition: each state's market rate survey sets its own, and some states let providers decide. NDCP converts hourly rates to weekly at 40 hours and daily rates at 5 days.

3.4 Employment path

A geography's projected employment rate, by sex and age band, is the national projected rate times the state-to-national ACS rate ratio. The national series uses Current Population Survey (CPS) history. I project it forward with CBO's long-term labor force participation projections. The ratios use ACS two-year pools. After 2024, the ratios stay at their 2024 values.

I aggregate the path to four caregiver age groups by sex. Then I apply the growth since 2019 to each group's base-year employment rate. The base-year rate is the weighted mean of the group's employment-forest probabilities.

4 State simulation

Each geography is a separate market; workers and families do not move between geographies. I retain the national equilibrium algorithm and annual demographic and income updates, with local wage growth and employment targets as described above. Group-specific employment shifts are solved in the baseline and held fixed across policy scenarios. The care-worker labor-supply elasticities remain 4 without a bachelor's degree and 2 with one. Non-labor residuals grow with the CPI.

5 Implementation and validation

1. **Demand.** The demand build passes for all 52 geographies. The 19 October check recovered p_0 after repricing and matched care hours in every geography, with a largest relative gap below 10^{-8}. Its 2026–2027 baseline also completed. These checks preceded the latest wage and staffing updates.

   Summing local targets does not reproduce national calibration totals, as reported above. One additional diagnostic fails: NY minus NYC unpaid-center hours differ from a separately built rest-of-state target by 1.22%, or 0.163% of total rest-of-state hours. The estimated child counts match; the entire 1.04 million annual-hour difference comes from the separate part-time mixes. I retain the local targets and report the discrepancy; the diagnostic tolerance is unchanged.

2. **Prices and wages.** The NDCP comparison uses care-mix weights from the price-adjusted calibration. Annual wage paths are implemented; all 52 saved paths match the source panel, and the 2019 source values reproduce the existing extract. The corrected full 52-geography, 1% sample baseline completed for 2026–2027 on 22 October, using the inputs from 227 reviewed staffing records. All 104 geography-year price solves converged with optimizer code zero; the largest paid-sector residual gap was \$0.000304 per child-hour, below the \$0.01 tolerance. All employment targets met the 10^{-5} tolerance in rate units; the largest gap was 0.00025 percentage points. Expanding the employment-shift search bounds resolved the earlier Florida

miss. Regression checks pass. This is a sample validation, not a full-sample production run; the latest staffing additions still require a new check.

The program-count source check passes locally and on the cluster after restoring its PDF dependency. The NY/NYC diagnostic above still fails at its original tolerance; I retain its explained difference rather than treating it as an exact accounting identity.

3. **Staffing.** I adopt 263 of 312 unweighted audit records, including 57 conditional mappings. The remaining 49 retain documented fallbacks: 40 center age points and 9 home records. Each geography has five center age points and one home record; these are records, not counts of children or providers. I still need historical evidence or a defensible category mapping for the fallbacks. New home capacities include age-composition and license-class limits; the supply inputs and price comparisons must be rebuilt as records are adopted.

5.1 Open items

1. **Historical staffing evidence.** Resolve the 49 remaining audit records where dated rules and comparable categories permit. Until then, retain the documented fallbacks. Record age limits, assistant requirements, and any category assumption alongside each adopted value.

2. **Updated-input validation.** Rebuild supply inputs after the staffing review, refresh the price comparison, and rerun the 52-geography baseline. Check employment targets alongside equilibrium convergence; distinguish the 1% sample check from a full-sample production run.

Source URLs and vintages are in `SOURCES.csv`; the staffing audit records legal alternatives, assumptions, effective dates, and source locators. Numerical consistency does not establish that care patterns or policy responses are empirically correct.